

Power BI Developer – Take-Home / Live Task

Manufacturing Quality: "Why are machines failing, and what should we watch?"

Time box: ~3–4 hours (take-home) or a 90-minute live build with the data pre-loaded. **Deliverable:** one .pbix file + a 3–5 minute walkthrough (recorded or live). **You may use:** any Power BI feature, AI tools, Google. We care about the result and your reasoning, not whether you memorized DAX.

The scenario

You've joined a manufacturing analytics team. A plant runs a fleet of machines that occasionally fail in production. Quality leadership doesn't want another table of numbers – they want a dashboard that **explains, in plain language, why failures happen and what to monitor**. Your audience is a mix of plant engineers (who know the machines) and managers (who don't). Both have to "get it" in 30 seconds.

Your job is **not** to build a predictive model. It is to **make a complex, multi-variable, imbalanced dataset understandable** to someone seeing it for the first time.

The data

AI4I 2020 Predictive Maintenance Dataset – public, UCI Machine Learning Repository.

- Download (CSV, ~10,000 rows, single file): <https://archive.ics.uci.edu/dataset/601/ai4i+2020+predictive+maintenance+dataset>
- Direct CSV often mirrored as ai4i2020.csv .

Columns:

Column	Meaning
UDI , Product ID	Row / unit identifiers
Type	Product quality variant: L (low, ~50%), M (medium, ~30%), H (high, ~20%)
Air temperature [K]	Ambient air temperature
Process temperature [K]	Machine process temperature
Rotational speed [rpm]	Spindle speed
Torque [Nm]	Applied torque
Tool wear [min]	Cumulative minutes the tool has been in use
Machine failure	1 = this unit failed (the headline outcome)
TWF	Tool-wear failure
HDF	Heat-dissipation failure
PWF	Power failure
OSF	Overstrain failure
RNF	Random failure

The five flags (TWF...RNF) are the **failure modes** – the *kind* of failure. A unit can in principle trip more than one. Treat Machine failure as the headline and the five modes as the breakdown.

You are encouraged to **derive new fields** where they make the story clearer.

What the dashboard must answer

Design however you like, but a viewer must be able to answer these without you narrating:

- How often do we fail, and is that a lot or a little?** Put the failure rate in context (it is small – make that honest and clear, don't bury it or exaggerate it).
- Which product type and which operating conditions fail most?** Let the viewer compare L / M / H and see how failures relate to the process variables.
- Why do failures happen?** Break failures down by mode (TWF/HDF/PWF/OSF/RNF) and show what operating conditions characterize each mode. (Hint: the modes do **not** all share the same driver – discovering and *showing* that difference is the heart of the task.)
- What should an engineer monitor?** End with a clear, defensible "watch these conditions" takeaway – a threshold, a danger zone, a rule of thumb – grounded in

what the data shows.

The dashboard should be **interactive**: a viewer should be able to filter to a product type or a failure mode and have the story update coherently.

Two things we deliberately left in the data

Strong candidates notice these without being told. We mention them so the take-home is fair:

- **Severe class imbalance** (~3–4% of units fail). A naive "% by category" chart can be misleading at this rate. Show that you can present rare events truthfully (rates with denominators/counts, not just proportions; avoid charts that imply false precision on tiny slices).
- **The labels don't perfectly reconcile**. The headline `Machine failure` flag and the OR of the five mode flags disagree on a small number of rows. We're interested in whether you **find it, surface it, and decide how to handle it** — not in a particular "right" answer.

Evaluation rubric (100 pts)

Dimension	Pts	What we're looking for
Clarity of explanation	30	A non-expert understands the story in 30s. Titles state findings, not field names ("Heat-dissipation failures cluster at low rpm + high temp gap" beats "HDF by RPM"). Right chart for each question.
Insight & data understanding	25	You found <i>why</i> the modes differ; you caught the imbalance and the label mismatch; you didn't over-claim.
Data modeling	15	Sensible model — derived columns/measures (e.g., temp gap, power $\approx$ torque $\times$ speed, wear buckets), a clean measure for failure <i>rate</i> , mode-level structure. Not everything jammed into one flat table.
Visual design & interactivity	15	Clean layout, consistent color, working slicers/cross-filtering or drill-through, readable on one screen.
DAX / transformation correctness	10	Rate measures are correct at every filter level; Power Query steps are tidy and documented.
Communication	5	The 3–5 min walkthrough is structured and honest about limitations.

Bonus (up to +10): a derived "danger zone" visual that combines two+ process variables; an uncertainty-aware treatment of rare modes (e.g., flagging low-n cells); a single executive summary card that someone could screenshot into a status email.

Practical notes

- Submit the `.pbix` plus a short note (½ page) on: your data model, any assumptions, what you'd add with more time, and one thing in the data you don't fully trust.
- If live: we'll have the CSV ready; spend the first 10 minutes exploring out loud before you build.
- There is no single correct dashboard. We are evaluating how you **turn complexity into clarity**.